

Using Single-Row Functions to Customize the Output

1. Write a query to display the current date. Label the column **Date**.

```
SQL> SELECT SYSDATE AS "Date" FROM DUAL;

Date
-----
19-JUL-26
```

2. The HR department needs a report to display the employee number, last name, salary, and salary increased by 15.5% (expressed as a whole number) for each employee. Label the column **New Salary**.

```
SQL> SELECT EMPNO, ENAME, SAL, ROUND(SAL * 1.155) NEW_SALARY FROM EMP;

EMPNO ENAME                      SAL NEW_SALARY
-----
7839 KING                        5000      5775
7698 BLAKE                      2850      3292
7782 CLARK                      2450      2830
7566 JONES                      2975      3436
7788 SCOTT                      3000      3465
7902 FORD                      3000      3465
7369 SMITH                      800       924
7499 ALLEN                     1600     1848
7521 WARD                      1250     1444
7654 MARTIN                    1250     1444
7844 TURNER                    1500     1733
7876 ADAMS                     1100     1271
7900 JAMES                      950      1097
7934 MILLER                    1300     1502

14 rows selected.
```

3. Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column **Increase**.

```
SQL> SELECT EMPNO, ENAME, SAL, ROUND(SAL*1.155) "New Salary", ROUND(SAL*1.155)-SAL "Increase" FROM EMP;

EMPNO ENAME                      SAL New Salary Increase
-----
7839 KING                        5000      5775      775
7698 BLAKE                      2850      3292      442
7782 CLARK                      2450      2830      380
7566 JONES                      2975      3436      461
7788 SCOTT                      3000      3465      465
7902 FORD                      3000      3465      465
7369 SMITH                      800       924      124
7499 ALLEN                     1600     1848      248
7521 WARD                      1250     1444      194
7654 MARTIN                    1250     1444      194
7844 TURNER                    1500     1733      233
7876 ADAMS                     1100     1271      171
7900 JAMES                      950      1097      147
7934 MILLER                    1300     1502      202

14 rows selected.
```

4. Write a query that displays the **ename** (with the first letter uppercase and all other letters lowercase) and the **length** of the ename for all employees whose name starts with the letters **J**, **A**, or **M**. Give each column an appropriate label. Sort the results by the employees' enames.

```
SQL> SELECT INITCAP(LOWER(ENAME)) "Employee Name", LENGTH(ENAME) "Name Length" FROM EMP WHERE ENAME LIKE 'J%' OR ENAME LIKE 'A%' OR ENAME LIKE 'M%' ORDER BY ENAME;
```

Employee Name	Name Length
Adams	5
Allen	5
James	5
Jones	5
Martin	6
Miller	6

6 rows selected.

5. Rewrite the query so that the user is prompted to enter a letter that starts the last name. For example, if the user enters **H** when prompted for a letter, then the output should show all employees whose last name starts with the letter **H**.

```
SQL> SELECT INITCAP(LOWER(ENAME)) "Employee Name", LENGTH(ENAME) "Name Length" FROM EMP WHERE ENAME LIKE UPPER('&LETTER')||'%' ORDER BY ENAME;
```

Enter value for letter: A

```
old 1: SELECT INITCAP(LOWER(ENAME)) "Employee Name", LENGTH(ENAME) "Name Length" FROM EMP WHERE ENAME LIKE UPPER('&LETTER')||'%' ORDER BY ENAME
new 1: SELECT INITCAP(LOWER(ENAME)) "Employee Name", LENGTH(ENAME) "Name Length" FROM EMP WHERE ENAME LIKE UPPER('A')||'%' ORDER BY ENAME
```

Employee Name	Name Length
Adams	5
Allen	5

6. The HR department wants to find the length of employment for each employee. For each employee, display the **ename** and calculate the number of months between today and the date on which the employee was hired. Label the column **MONTHS_WORKED**. Order your results by the number of months employed. Round the number of months up to the closest whole number.

```
SQL> SELECT ENAME, CEIL(MONTHS_BETWEEN(SYSDATE, HIREDATE)) AS MONTHS_WORKED FROM EMP ORDER BY MONTHS_WORKED;
```

ENAME	MONTHS_WORKED
SCOTT	469
ADAMS	470
MILLER	534
FORD	536
JAMES	536
KING	537
MARTIN	538
TURNER	539
CLARK	542
BLAKE	543
JONES	544
WARD	545
ALLEN	545
SMITH	548

14 rows selected.

7. Create a report that produces the following for each employee: earns monthly salary but wants **3 times salary**. Label the column **Dream Salaries**.

```
SQL> SELECT ENAME, SAL, SAL*3 AS "Dream Salaries" FROM EMP;
```

ENAME	SAL	Dream Salaries
KING	5000	15000
BLAKE	2850	8550
CLARK	2450	7350
JONES	2975	8925
SCOTT	3000	9000
FORD	3000	9000
SMITH	800	2400
ALLEN	1600	4800
WARD	1250	3750
MARTIN	1250	3750
TURNER	1500	4500
ADAMS	1100	3300
JAMES	950	2850
MILLER	1300	3900

14 rows selected.

8. Create a query to display the last name and salary for all employees. Format the salary to be **15 characters long, left-padded with the \$ symbol**. Label the column **SALARY**.

```
SQL> SELECT ENAME, LPAD(SAL,15,'$') AS SALARY FROM EMP;
```

ENAME	SALARY
KING	\$\$\$\$\$\$\$\$\$\$\$\$5000
BLAKE	\$\$\$\$\$\$\$\$\$\$\$\$2850
CLARK	\$\$\$\$\$\$\$\$\$\$\$\$2450
JONES	\$\$\$\$\$\$\$\$\$\$\$\$2975
SCOTT	\$\$\$\$\$\$\$\$\$\$\$\$3000
FORD	\$\$\$\$\$\$\$\$\$\$\$\$3000
SMITH	\$\$\$\$\$\$\$\$\$\$\$\$800
ALLEN	\$\$\$\$\$\$\$\$\$\$\$\$1600
WARD	\$\$\$\$\$\$\$\$\$\$\$\$1250
MARTIN	\$\$\$\$\$\$\$\$\$\$\$\$1250
TURNER	\$\$\$\$\$\$\$\$\$\$\$\$1500
ADAMS	\$\$\$\$\$\$\$\$\$\$\$\$1100
JAMES	\$\$\$\$\$\$\$\$\$\$\$\$950
MILLER	\$\$\$\$\$\$\$\$\$\$\$\$1300

14 rows selected.

9. Display each employee's last name, hire date, and salary review date, which is the **first Monday after six months of service**. Label the column **REVIEW**. Format the dates to appear in the format similar to **"Monday, the Thirty-First of July, 2000."**

```
SQL> SELECT ENAME, TO_CHAR(HIREDATE,'fmDay, "the" DDSPTH "of" Month, YYYY') HIREDATE, TO_CHAR(NEXT_DAY(ADD_MONTHS(HIREDATE,6),'MONDAY'), 'fmDay, "the" DDSPTH "of" Month, YYYY') REVIEW FROM EMP;
```

ENAME	HIREDATE	REVIEW
KING	Monday, the TWENTY-FOURTH of May, 1982	Tuesday, the SEVENTEENTH of November, 1981
BLAKE	Monday, the SECOND of November, 1981	Friday, the FIRST of May, 1981
CLARK	Monday, the FOURTEENTH of December, 1981	Tuesday, the NINTH of June, 1981
JONES	Monday, the FIFTH of October, 1981	Thursday, the SECOND of April, 1981
SCOTT	Monday, the EIGHTEENTH of January, 1988	Monday, the THIRTEENTH of July, 1987
FORD	Monday, the SEVENTH of June, 1982	Thursday, the THIRD of December, 1981
SMITH	Monday, the TWENTY-SECOND of June, 1981	Wednesday, the SEVENTEENTH of December, 1980
ALLEN	Monday, the TWENTY-FOURTH of August, 1981	Friday, the TWENTIETH of February, 1981
WARD	Monday, the TWENTY-FOURTH of August, 1981	Sunday, the TWENTY-SECOND of February, 1981
MARTIN		Monday, the TWENTY-EIGHTH of September, 1981

10. Display the last name, hire date, and day of the week on which the employee started. Label the column **DAY**. Order the results by the day of the week, starting with **Monday**.

```
SQL> SELECT ENAME, HIREDATE, TO_CHAR(HIREDATE,'DAY') AS DAY FROM EMP ORDER BY TO_CHAR(HIREDATE,'D');
```

ENAME	HIREDATE	DAY
WARD	22-FEB-81	SUNDAY
MARTIN	28-SEP-81	MONDAY
SCOTT	13-JUL-87	MONDAY
KING	17-NOV-81	TUESDAY
TURNER	08-SEP-81	TUESDAY
CLARK	09-JUN-81	TUESDAY
SMITH	17-DEC-80	WEDNESDAY
FORD	03-DEC-81	THURSDAY
JONES	02-APR-81	THURSDAY
JAMES	03-DEC-81	THURSDAY
ALLEN	20-FEB-81	FRIDAY
BLAKE	01-MAY-81	FRIDAY
ADAMS	23-MAY-87	SATURDAY
MILLER	23-JAN-82	SATURDAY

14 rows selected.

11. Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show "**No Commission.**" Label the column **COMM**.

```
SQL> SELECT ENAME, NVL(TO_CHAR(COMM),'No Commission') AS COMM FROM EMP;
```

ENAME	COMM
KING	No Commission
BLAKE	No Commission
CLARK	No Commission
JONES	No Commission
SCOTT	No Commission
FORD	No Commission
SMITH	No Commission
ALLEN	300
WARD	500
MARTIN	1400
TURNER	0
ADAMS	No Commission
JAMES	No Commission
MILLER	No Commission

14 rows selected.

12. Create a query that displays the first **eight characters** of the employees' last names and indicates the amounts of their salaries with **asterisks**. Each asterisk signifies a **thousand dollars**. Sort the data in descending order of salary. Label the column **EMPLOYEES_AND_THEIR_SALARIES**.

```
SQL> SELECT SUBSTR(ENAME,1,8)||' '||RPAD('*',SAL/1000,'*') AS EMPLOYEES_AND_THEIR_SALARIES FROM EMP ORDER BY SAL DESC;
EMPLOYEES_AND_THEIR_SALARIES
-----
KING *****
SCOTT ***
FORD ***
JONES **
BLAKE **
CLARK **
ALLEN *
TURNER *
MILLER *
MARTIN *
WARD *
ADAMS *
JAMES
SMITH
14 rows selected.
```

13. Using the **DECODE** function, write a query that displays the grade of all employees based on the value of the column **JOB**, using the following data:

- PRESIDENT → A
- MANAGER → B
- SALESMAN → C
- CLERK → D

```
SQL> SELECT ENAME, JOB, DECODE(JOB,'PRESIDENT','A','MANAGER','B','SALESMAN','C','CLERK','D','ANALYST','E') AS GRADE FROM EMP;
ENAME                                JOB                                GRADE
-----                                -
KING                                PRESIDENT                          A
BLAKE                                MANAGER                            B
CLARK                                MANAGER                            B
JONES                                MANAGER                            B
SCOTT                                ANALYST                            E
FORD                                ANALYST                            E
SMITH                                CLERK                              D
ALLEN                                SALESMAN                           C
WARD                                SALESMAN                           C
MARTIN                              SALESMAN                           C
TURNER                              SALESMAN                           C
ADAMS                                CLERK                              D
JAMES                                CLERK                              D
MILLER                              CLERK                              D
14 rows selected.
```

14. Rewrite the statement in the preceding exercise using the **CASE** syntax.

```
SQL> SELECT ENAME, JOB, CASE JOB WHEN 'PRESIDENT' THEN 'A' WHEN 'MANAGER' THEN 'B' WHEN 'SALESMAN' THEN 'C' WHEN 'CLERK' THEN 'D' END  
AS GRADE FROM EMP;
```

ENAME	JOB	GRADE
KING	PRESIDENT	A
BLAKE	MANAGER	B
CLARK	MANAGER	B
JONES	MANAGER	B
SCOTT	ANALYST	
FORD	ANALYST	
SMITH	CLERK	D
ALLEN	SALESMAN	C
WARD	SALESMAN	C
MARTIN	SALESMAN	C
TURNER	SALESMAN	C
ADAMS	CLERK	D
JAMES	CLERK	D
MILLER	CLERK	D

14 rows selected.